

Poker Math

HCI 520 Term Project Design Specifications

Topic Description:

My term project learning system will focus on the topic of **Poker Math**. Specifically, the system will teach the concepts of hand “equity” and pot odds, giving learners a mathematical foundation for judging the profitability of a call in a poker game. Content information will come from Wikipedia pages on these widely understood mathematical concepts, blog and educational articles from poker-specific platforms like Upswing Poker or GTOWizard, and supplemented by my existing familiarity with the subject.

Learner Description:

Learners might primarily use this system to improve their poker play by establishing an understanding of the game’s mathematical foundations. Secondary motivations might be using the system as a transfer of learning or varied context practice exercise to reinforce learning about mathematical concepts of probability and risk/reward.

Before using the system, I would assume the learners would have an understanding of:

- A standard 52-card deck of cards
 - Recalling **[1.2]** the composition of the 52 cards
 - **[Bc Knowledge of theories, models, and structures]**
- The basic rules of Texas Hold ‘Em poker
 - Recalling **[1.2]** the rules of Texas Hold ‘Em
 - **[Bc Knowledge of theories, models, and structures]**

Learning Objectives:

LO1: Understand hand “equity” and how to estimate it

- Executing **(3.1)** the rule of 2-and-4 to quickly estimate the equity of any given hand
- **(Ca Knowledge of subject-specific skills and algorithms)**

LO2: Understand pot odds and how to calculate it

- Executing **(3.1)** the pot odds formula to calculate minimum required equity
- **(Ca Knowledge of subject-specific skills and algorithms)**

LO3: Use pot odds to judge the profitability of a call

- Judging **(5.2)** whether a call is profitable by comparing hand equity and the required equity threshold produced by pot odds
- **(Bc Knowledge of theories, models and structures)**

Assessment:

LO1: Hand Equity and the Rule of 2-and-4

- **Method:** This objective will be assessed by presenting the learner with the necessary elements required to execute the rule of 2-and-4 algorithm and calculate their hand's equity (e.g. a two-card hand and three-card flop). This will assess their ability to execute (3.1) with their knowledge of a subject-specific algorithm (Ca).
- **Example:** *You hold Ah and Kh. The flop shows Qh, 8h, and 7c. Use the rule of 2-and-4 to estimate your hand's equity.*

LO2: Pot Odds Calculation

- **Method:** This objective will be assessed by presenting the learner with the necessary factors required to execute a pot odds calculation (e.g. the size of the pot and cost of the call). This will assess their ability to execute (3.1) with their knowledge of a subject-specific algorithm (Ca).
- **Example:** *There is \$40 in the pot. Your opponent bets \$10. Calculate your pot odds as a ratio, and then convert it into a percentage.*

LO3: Judging Call Profitability

- **Method:** This objective will be assessed by presenting the learner with the necessary factors required to judge a call's profitability (e.g. all factors used in previous LOs). This will assess their ability to judge (5.2) a call using their knowledge of the mathematical models of poker (Bc).
- **Example:** *You hold Ah and Kh. The flop shows Qh, 8h, and 7c. There is \$40 in the pot. Your opponent bets all-in for \$40. Calculate your equity and pot odds, then compare them to determine if calling is profitable.*

Teaching:

Topic Outline

1. Introduction to Poker

- a. Before jumping into the learning objectives, there will be a quick preliminary section establishing the baseline poker knowledge learners should have before using the system. If unfamiliar, the learner will also be directed to cheat sheets in a dropdown menu where they can refresh their knowledge of the cards/structure of a 52-card deck, the ranking of different winning poker hands, and the basic rules and jargon of Texas Hold 'Em. Learners will also be taught commonly used poker hand notation to streamline system communication (e.g. Ace of hearts = Ah, 10 of clubs = Tc etc.).

2. LO1: Hand Equity and the Rule of 2-and-4

- a. In this section, learners will be taught the mathematical concept of “equity” in poker, or in other words, the chance a player has to win the hand at “showdown” (when all the cards are revealed).
- b. Learners will then be presented with the actual equity calculation using the addition law of probability to demonstrate the underlying math.
- c. Learners will then be taught the Rule of 2-and-4, a mathematical shorthand that approximates the previous actual equity calculation that relies on a 52-card deck being almost exactly one half of 100, a required factor in any percentage calculation.
- d. Assessment: See method and example question in respective LO of previous section.

3. LO2: Pot Odds Calculation

- a. In this section, learners will be taught the mathematical concept of “pot odds” in poker, or in other words, the ratio of the current size of the pot to the cost of the considered call.
- b. Learners will then be presented with the concept within the context of a poker game, demonstrating how the odds can fluctuate depending on how many players have bet or called, how much money they have risked, and otherwise connect the abstract mathematical concept to actual poker actions.

- c. Assessment: See method and example question in respective LO of previous section.

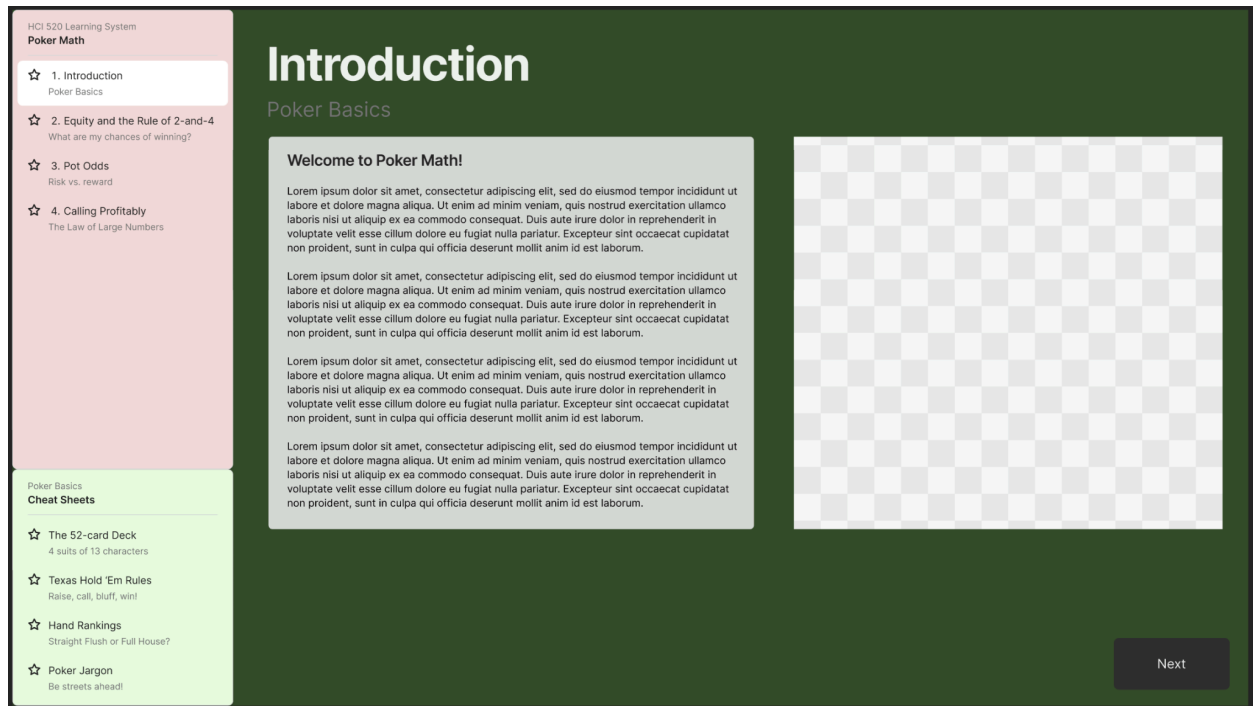
4. LO3: Judging Call Profitability

- a. In this section, learners will be taught how to connect the concepts learned in previous learning objectives, and use them to judge whether a call is profitable or not in the long-term.
- b. Learners will then be presented with the law of large numbers and common illustrative thought experiments or “toy games” to demonstrate the concept at-large. (e.g. “Would you risk \$1 for a 50/50 chance at \$100?”)
- c. Learners will then be taught how to stack the previous concepts together to make profitability judgments by using pot odds to find the required equity to call, and comparing against the calculated equity of your own hand.
- d. Assessment: See method and example question in respective LO of previous section.

Wireframes

Informational Screen:

Educational content presented as text, alongside contextual graphics:



Assessment Screen:

Input fields with a “Check Answer” button, alongside graphic elements required to complete:

HCI 520 Learning System
Poker Math

☆ 1. Introduction
Poker Basics

☆ 2. Equity and the Rule of 2-and-4
What are my chances of winning?

☆ 3. Pot Odds
Risk vs. reward

☆ 4. Calling Profitably
The Law of Large Numbers

Poker Basics
Cheat Sheets

☆ The 52-card Deck
4 suits of 13 characters

☆ Texas Hold 'Em Rules
Raise, call, bluff, win!

☆ Hand Rankings
Straight Flush or Full House?

☆ Poker Jargon
Be streets ahead!

Equity and the Rule of 2-and-4

Assessment

Your opponent goes all-in on the flop.
Use the Rule of 2-and-4 to calculate your equity:

of "outs":

of remaining streets:

Your hand's equity:

The Board:

♥ Q

♥ 8

♦ 7

Your Hand:

♥ A

♥ K